

Milestone Report of Let's Make Pottery

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1 Introduction

In archaeological excavations, damaged artifacts are often unearthed, and restoring these artifacts can be both challenging and time-consuming for archaeologists. In our project, we aim to develop a deep learning-based generative model to assist archaeologists in efficiently and accurately restoring artifact fragments.

2 Problem Statement

We model this task as training a deep learning model that takes one or more fragments of any type of pottery as input and outputs voxel data of a complete pottery piece that closely matches its true shape. The goal is to enable the restoration of arbitrary artifact fragments.

The data sets we use consist of .vox files generated from real pottery models, comprising 11 different types. These files adhere to the MagicalVoxel format, where each file may contain fragments labeled with positive integers, and 0 indicates an empty space. To evaluate the similarity between the model's output and the true shape of the pottery, we employ the Dice Similarity Coefficient and visualize the differences in shape for better analysis.

3 Technical Approach

Our plan begins with visualizing the voxel data using the provided code framework to ensure its correctness. Next, we design a specialized Dataset class for voxel data to standardize neural network inputs. Based on this foundation, we implement an AE-GAN model to learn a mapping function between two distributions (fragments and complete pottery), enabling the generation of plausible restoration results from input fragments.

To better understand the .vox files in the data sets and ensure they are fed into the neural network as expected, we developed methods for visualizing these data (details provided in the Preliminary Result).

We redefined the Dataset loading class, inheriting from torch.utils.data.Dataset, to facilitate the retrieval of voxel data during training and testing. A major challenge of this task lies in designing an appropriate GAN model. For this, we plan to adopt a network structure similar to UNet.

For evaluating the similarity between generated and real pottery, we use the Dice Similarity Coefficient (DSC) to quantitatively measure their resemblance. A value closer to 1 indicates higher similarity. Additionally, we plan to leverage the Plotly library to visualize the final outputs.

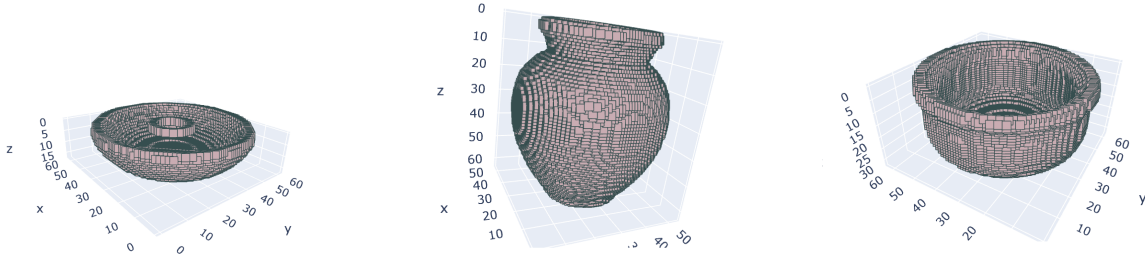


Figure 1: Complete Pottery

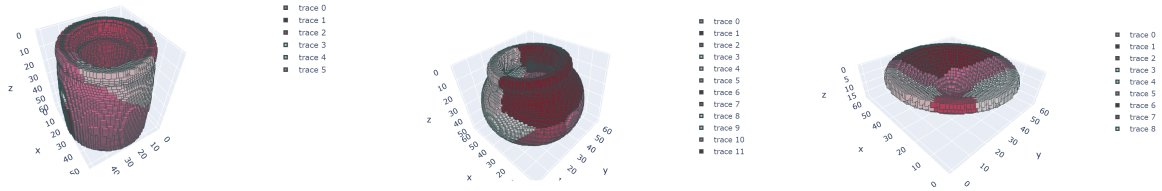


Figure 2: Visualize Fragments of Pottery

4 Preliminary Result

We have successfully visualized the data recorded in .vox files and conducted unit testing using Jupyter. The system can now visualize a complete pottery piece (Figure 1) and show which fragments it is composed of (Figure 2). Additionally, users can interactively add or remove fragments to observe how the pottery changes (Figure 3).

Moreover, we have rewritten the Dataset class to enable efficient data extraction at a given resolution. In training mode, the Dataset calls `__getitem__()` to select a sample and randomly pick one of its fragments. In testing mode, the Dataset calls `__getitem__specific_frag__()`, which is similar to `__getitem__()`, but it selects a fixed fragment.

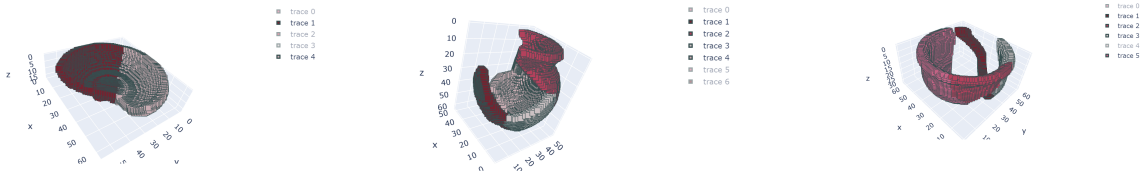


Figure 3: Interactively Check Fragments